

1.1
1

- a. Yes, True
- b. Yes, False
- c. Yes, True
- d. Yes, False
- e. No (contains a variable)
- f. No (this is a command)

1.1
5

- a. Mei does not have an MP3 player
- b. There is pollution in New Jersey
- c. $2 + 1 \neq 3$
- d. The summer in Maine is not hot or not sunny
(De Morgan's Law)

1.1
7

- a. Steve does not have more than 100 GB free disk space on his laptop
- b. Zack doesn't block emails from Jennifer, or Zack doesn't block texts from Jennifer (De Morgan's Law)
- c. $7 \cdot 11 \cdot 13 \neq 999$
- d. Diana did not ride her bicycle 100 miles on Saturday

1.1
8

Smartphone	RAM	ROM	Camera
A	256 MB	32 GB	8 MP
B	288 MB	64 GB	4 MP
C	128 MB	32 GB	5 MP

- a. True ($288 > 256$, and $288 > 128$)
- b. True (C has a higher resolution camera)
- c. False (B does not have a higher resolution camera than A)
- d. False (Hypothesis is true but conclusion is false.)
- e. False (One side of biconditional is true the other side is false)

Company	Annual Revenue	Net Profit
Acme Computer	138 billion dollars	8 billion dollars
Nadir Software	87 billion dollars	5 billion dollars
Quixote Media	111 billion dollars	13 billion dollars

- False (Acme has higher revenue)
- True (both parts are true, so the "and" statement is true)
- True (Quixote had most profit, so the "or" statement is true)
- True (Hypothesis is false, so the statement must be true)
- True (Both parts of the biconditional have the same truth value)

1.1
11h "Swimming at the New Jersey shore is not allowed, and either swimming at the New Jersey shore is allowed, or sharks have not been spotted near the shore."

("either .. or" replicates the parentheses in the original statement)

- 1.1
13
- $p \wedge q$ p : "It is below freezing"
 - $p \wedge \neg q$ q : "It is snowing"
 - $\neg p \wedge \neg q$
 - $p \vee q$
 - $p \rightarrow q$
 - $(p \vee q) \wedge (p \rightarrow \neg q)$
 - $p \longleftrightarrow q$

- 1.1
16d
- p : "You got an A on the final"
 - q : "You do every exercise in this book"
 - r : "You get an A in this class"

$$p \wedge \neg q \wedge r$$

- 1.1
17
- p: "Grizzly bears have been seen in the area"
 q: "Hiking is safe on the trail"
 r: "Berries are ripe along the trail"

a. $r \wedge \neg p$

b. $\neg p \wedge q \wedge r$

c. $r \rightarrow (q \leftrightarrow \neg p)$

d. $\neg q \wedge \neg p \wedge r$

e. $(q \rightarrow (\neg r \wedge p)) \wedge \neg((\neg r \wedge \neg p) \rightarrow q)$

f. $(p \wedge r) \rightarrow \neg q$

- 1.1
19
- a. False, hypothesis is true but conclusion is false
 b. True, hypothesis is false
 c. True, hypothesis is false
 d. True, hypothesis is false

1.1
25h

If the water is not too cold, then Jan will go swimming.
 ("if p, then q" $\leftrightarrow$ "q unless $\neg p$ ")

1.1
29a

If it snows today, I will ski tomorrow.
 converse: If I ski tomorrow, then it snowed today
 contrapositive: If I do not ski tomorrow, then it did not snow today.
 inverse: If it did not snow today, then I will not ski tomorrow.

1.1
30c

When I stay up late, it is necessary that I sleep until noon.
 If I stay up late, then I will sleep until noon.

converse: If I sleep until noon, then I will stay up late.

contrapositive: If I don't sleep until noon, then I don't stay up late.

inverse: If I don't stay up late, then I don't sleep until noon.

1.1 32c unique variables: $p, r, s, t, u, v \Rightarrow 2^6 = 64$ rows

1.1 33d

p	q	$p \vee q$	$p \wedge q$	$(p \vee q) \rightarrow (p \wedge q)$
T	T	T	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	F	T

1.1 35f

p	q	$p \oplus q$	$\neg q$	$p \oplus \neg q$	$(p \oplus q) \rightarrow (p \oplus \neg q)$
T	T	F	F	T	T
T	F	T	T	F	F
F	T	T	F	F	F
F	F	F	T	T	T

1.1 37a

p	q	$\neg q$	$p \rightarrow \neg q$
T	T	F	F
T	F	T	T
F	T	F	T
F	F	T	T

1.1 39a

p	q	r	$\neg q$	$\neg q \vee r$	$p \rightarrow (\neg q \vee r)$
T	T	T	F	T	T
T	T	F	F	F	F
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	F	T	T
F	T	F	F	F	F
F	F	T	T	T	T
F	F	F	T	T	T

1.1 48a $11000 \wedge (01011 \vee 11011)$

$$\begin{array}{r} 01011 \\ \vee 11011 \\ \hline 11011 \end{array}$$

$$\begin{array}{r} 11000 \\ \wedge 11011 \\ \hline 11000 \end{array}$$

(1.2)
1 e: "You can edit a protected Wikipedia entry"

a: "You are an administrator"

You cannot edit a protected Wikipedia entry unless you are an administrator.
($\neg e$ unless a)

$$\neg a \rightarrow \neg e$$

(1.2)
3 g: "You can graduate"

m: "You owe money to the university"

r: "You have completed the requirements of your major"

b: "You have an overdue library book"

$$g \rightarrow (r \wedge \neg m \wedge \neg b)$$

(1.2)
5 e: "You are eligible to be President of the U.S.A."

a: "You are at least 35 years old"

b: "You were born in the USA"

p: "At the time of your birth, both of your parents were citizens"

r: "You have lived at least 14 years in the U.S.A."

$$e \rightarrow (a \wedge (b \vee p) \wedge r)$$

(1.2)
11 S: "The router can send packets to the edge system"

a: "The router supports the new address space"

r: "The latest software release is installed"

$$S \rightarrow a, \quad a \rightarrow r, \quad r \rightarrow S, \quad \neg a \text{ are all true.}$$

Since we know $\neg a$ is true, a must be false.

Since we know a is false and $S \rightarrow a$ is true, S must be false.

Since we know S is false and $r \rightarrow S$ is true, r must be false.

Since r is false, S is false, $r \rightarrow S$ is still true.

Thus we have found no contradictions, so the system is consistent.

(1.2)
17

Trunk 1: "The treasure is in Trunk 3"

Trunk 2: "The treasure is in Trunk 1"

Trunk 3: "This trunk is empty"

a.) "All inscriptions are false"

Not possible, because then Trunk 1 would claim "the treasure is not in Trunk 3" while Trunk 3 would claim "This trunk is not empty," a contradiction.

b.) "Exactly one of these inscriptions is true."

Possible, but we can't tell where the treasure is.

If Trunk 1 is true, the treasure is in Trunk 3.

If Trunk 2 is true, we have a contradiction.

If Trunk 3 is true, the treasure is in Trunk 2.

So the treasure could be in either Trunk 2 or Trunk 3.

c.) "Exactly two of these inscriptions are true" (aka exactly one is false)

Possible, and the treasure is in Trunk 1.

If Trunk 1 is false, the treasure is in Trunk 1.

If Trunk 2 is false, we have a contradiction.

If Trunk 3 is false, we have a contradiction.

So Trunk 2 and 3 must be true with the treasure in Trunk 1.

d.) "All three inscriptions are true."

Not possible, the treasure can't be in both Trunk 1 and Trunk 3 so we have a contradiction.

1.2
30) No solution, neither the knight nor the knave could claim that they are the knave.

1.2
31) B can't be the knight because if they were telling the truth about A then there would be two knights.

C can't be the knight because they claim that they are not.

Thus A is the knight.

Thus B was telling the truth, so B must be the spy and C must be the knave.

1.2
35) No solution, because whoever the knave is would be telling the truth, a contradiction.

1.2
39) b: the butler is telling the truth

c: the cook is telling the truth

g: the gardener is telling the truth

h: the handyman is telling the truth

$$b \rightarrow c$$

$$\neg(c \wedge g) \equiv \neg c \vee \neg g$$

$$\neg(\neg g \wedge \neg h) \equiv g \vee h$$

$$h \rightarrow \neg c$$

Suppose b is true. Then c is true, so g is false, so h is true.

But $h \rightarrow \neg c$ becomes $T \rightarrow F \equiv F$, a contradiction.

So b must be false.

If c were true, by the same argument we would have a contradiction.

So c must be false.

So the butler and the cook are both lying.

We can't know anything further about g or h.

